

GLYZEL GALAGAR

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PROFESSIONAL SUMMARY

Backend-focused full-stack developer skilled in Java, Spring Boot, Python, FastAPI, PostgreSQL, AWS, and AI engineering. Experienced in building end-to-end software systems, including SaaS platforms, RAG applications, AI agents, adaptive learning systems, and secure REST APIs. Strong in backend development, AI integration, databases, cloud infrastructure, and security.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, C#, SQL, HTML, CSS

Backend Development: Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, REST API design, OAuth2, JWT, MapStruct, RabbitMQ, Microservices, Modular Monolith

Frontend Development: React.js, Tailwind CSS, React Router, Axios, Responsive Design, Design Systems

Artificial Intelligence: LangChain, LangGraph, LangChain4j, Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant, pgvector), LLM Tool Calling, Prompt Engineering, Embeddings, Document Ingestion, Bayesian Knowledge Tracing

Cloud and DevOps: AWS (EC2, S3, RDS, Cognito, Lambda), Vercel, Hostinger, Docker, Docker Compose, Railway, Git, GitHub, CI/CD

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Database Design, ERD Modeling, SQL Optimization

Tools and Practices: IntelliJ IDEA, Postman, Maven, Agile, Software Development Life Cycle (SDLC), Unit and Integration Testing, Code Review, Multi-Tenant Architecture

Operating Systems: Linux, Windows

PROJECTS

REBYU - AI-Powered Review and Learning Management Platform | Capstone Project

Technologies: React.js, Spring Boot, FastAPI, PostgreSQL, AWS (Cognito, S3, RDS), LangChain, LangGraph, Qdrant, Docker, RabbitMQ, Vercel, Hostinger

- Designed a multi-tenant SaaS platform for IT certification review spanning 20+ bounded domains (certifications, assessments, enrollment, gamification, community, billing, enterprise management), with a modular Spring Boot core and a dedicated FastAPI service for AI and learning analytics.
- Engineered a modular Spring Boot backend with a dedicated FastAPI AI/analytics service, Spring Data JPA, transactional outbox processing, and projection-based queries.
- Built AI content pipelines on LangChain/LangGraph that generate full curricula, lessons, quizzes, mock exams, flashcards, and study plans — with per-task model routing across providers and strict JSON-schema validation, chosen after benchmarking which models could reliably sustain multi-turn tool calls.
- Developed an adaptive assessment and personalization engine supporting multiple question types, retakes, scoring, validation, and Bayesian Knowledge Tracing to estimate per-skill mastery and drive personalized recommendations.

INFRASTRACK - AI-Powered Construction Monitoring Platform

Technologies: React, TanStack Query, Tailwind CSS, Spring Boot, Spring Security, JPA, Hibernate, PostgreSQL, pgvector, LangChain4j, AWS (Cognito, S3), Docker, Vercel

- Built an AI-powered construction monitoring platform where clients can track project progress and ask questions about milestones, dates, costs, and plans through a natural-language assistant.
- Developed a LangChain4j RAG pipeline that ingests PDF, Word, Excel, PowerPoint, and text reports, converts them within the application, and stores searchable embeddings in PostgreSQL/pgvector without relying on paid document-processing services.
- Secured the platform with Amazon Cognito role-based access control and Spring Security OAuth2/JWT validation, integrated S3 presigned URLs for protected project files and photos, and built role-aware React dashboards and end-to-end project workflows.

INTERBYU - AI Interview Simulation Platform

Technologies: Python, FastAPI, LangGraph, LangChain, Groq, PostgreSQL, Qdrant, React, AWS Cognito, Vercel

- Built an AI platform that screens resumes against real job postings and runs voice-driven mock interviews with per-answer scoring and structured feedback.
- Orchestrated 8 LLM agents through LangGraph state machines with schema-validated structured outputs, plus RAG over resume embeddings in Qdrant to ground questions in real experience.

E-BOTO - University Election Management System

Technologies: C#, Windows Forms, Entity Framework, SQL Server

- Developed a desktop election management system for university-wide student elections, covering candidates, positions, voters, ballot casting, and automated result tallying.
- Designed the entity relationship diagram and relational database schema, and enforced one vote per student through unique constraints and validation logic.
- Implemented full create, read, update, and delete functionality using Entity Framework, and delivered an administrator interface in Windows Forms following the software development life cycle.

EDUCATION

University of Cebu Lapu-Lapu and Mandaue | Mandaue City, Cebu

Bachelor of Science in Information Technology, 2023 to Present

Senior High School, Science, Technology, Engineering, and Mathematics Strand, 2022 to 2023

AWARDS AND SCHOLARSHIPS

- Second Place, General Information Technology Quiz, 11th ICT Congress 2025, SM Seaside Cebu, April 2025
- Champion, General Information Technology Quiz, UCLM CCS Days 2025, Mandaue City, Cebu, February 2025
- Overseas Workers Welfare Administration Education for Development Scholarship Program Scholar, July 2023 to Present

TRAINING AND WORKSHOPS

- Blockchain Technology Workshop on Coding Smart Contracts in CashScript, Certificate of Participation, Mandaue City, Cebu, November 2025

ADDITIONAL SKILLS

Problem solving and debugging, collaborative team development, fast learning of emerging technologies, time management, attention to detail, clean and maintainable code practices.